Response to Reviewers — Fourth Revision

Submission ID: fbf130b1-76af-4698-aa61-7b55e8721359
Manuscript: A Four-Dimension Pre-Modeling Audit Protocol for Educational Prediction Benchmarks
Journal: Scientific Reports
Revision: Fourth round

Dear Editor and Reviewers,

We thank the editor and reviewers for their continued engagement with our manuscript. We are grateful to Reviewer 3 for the positive assessment and for confirming that the paper is now ready for acceptance; no further issues were raised by Reviewer 3 in this round. Reviewer 2 raised two further issues, both concerning the interpretation and scope of the feature-attribution analysis (Section 3.6). Both points are well-taken: the previous wording overstated the strength of the conclusions and generalized the feature-attribution findings beyond the three datasets actually analysed. We have revised the feature-attribution section accordingly, softening the interpretive claims to exploratory associations that are consistent with—but do not validate—the audit profiles, and restricting descriptive statements to the three analysed datasets (OULAD, UCI Student, Higher Ed, corresponding to Strong, Mostly Passing, and Partial profiles respectively).

The main audit framework, the four-dimension results, and all numerical analyses are unchanged. We describe each change below. Specific line references refer to the fourth-revision manuscript files.

Response to Reviewer 2

We are grateful to Reviewer 2 for this final round of careful reading. The two remaining concerns are addressed below.

Comment 1: Interpretation of the feature-attribution analysis

Reviewer: The interpretation of the feature-attribution analysis remains stronger than the evidence supports. Statements describing IID attribution instability as a "necessary condition" for cross-group fragility, claiming that it "confirms" model dependence, or presenting attribution stability as a "proxy for dataset readiness" should be revised. Since the analysis does not directly assess attribution stability under group-aware holdout, the findings should be presented as exploratory associations that are consistent with, but do not validate, the audit profiles.

Response: We thank the reviewer for this precise and well-founded observation. The reviewer is correct: because the feature-attribution analysis is conducted only under IID splits and does not directly evaluate attribution stability under group-aware holdout, the previous framing exceeded what the evidence supports. We have revised the entire feature-attribution section to present the findings as exploratory associations rather than validation of the audit profiles.

The specific changes are as follows.

1. **"Necessary condition" removed.** The previous claim that "Such lower-bound instability is a necessary—though not sufficient—condition for cross-group fragility" has been replaced with the explicit

scoping statement: "Because this analysis is conducted only under IID splits and does not directly assess attribution stability under group-aware holdout, the observed patterns provide exploratory associations that are consistent with—but do not validate—the audit profiles." (behavioraudit.tex, line 325)

2. **"Confirms" softened.** The previous claim that the cross-model comparison was "confirming that attribution rankings from fragile datasets are model-dependent as well as split-dependent" has been revised to "suggesting that attribution rankings from the Partial and Mostly Passing datasets are more model-dependent as well as split-dependent". The verb "confirm" has been replaced by "suggest", and the scope is now restricted to the analysed datasets. (behavioraudit.tex, line 325)

3. **"Proxy for dataset readiness" removed.** The previous concluding sentence stated that "iid-based attribution stability provides a proxy for dataset readiness: Strong datasets maintain stable attribution patterns across splits, while Fragile datasets exhibit split-dependent rankings." This claim has been removed. The revised conclusion reads: "OULAD (Strong) maintains stable attribution patterns across splits, while UCI Student (Mostly Passing) and Higher Ed (Partial) exhibit more split-dependent rankings. These exploratory associations do not, by themselves, establish a causal link between IID attribution instability and cross-group fragility." (behavioraudit.tex, line 325)

We have also re-read the rest of the manuscript (Methods, Results, Discussion, and Conclusion) and confirmed that the remaining uses of "confirm" appear in the group-holdout and train-test-gap analyses, where the verb is supported by direct numerical evidence; those uses are unchanged. The feature-attribution section now uses only exploratory language throughout.

Comment 2: Scope of the feature-attribution conclusions

Reviewer: The scope of the feature-attribution conclusions should be corrected. The analysis includes only OULAD, UCI Student, and Higher Ed, corresponding respectively to Strong, Mostly Passing, and Partial profiles; it does not include a dataset formally classified as Fragile. Therefore, statements referring generally to "Fragile datasets" or claiming a pattern extending from Fragile to Strong should be replaced with wording restricted to the three datasets actually analysed.

Response: We agree with the reviewer that the previous wording generalized the feature-attribution findings beyond the analysed scope. The feature-attribution analysis includes three datasets—OULAD (Strong), UCI Student (Mostly Passing), and Higher Ed (Partial)—and does not include a Fragile-profile dataset. We have corrected all statements in the feature-attribution section that referred to "Fragile datasets" or implied a Fragile-to-Strong continuum.

The specific changes are as follows.

1. **Opening scope statement.** The previous sentence stated that the three datasets were "chosen to span Fragile-to-Strong profiles". The revised text now reads: "These three datasets were chosen to span three of the four readiness profiles (Strong, Mostly Passing, and Partial); the Fragile profile (MM-TBA) is not included in this analysis because its small sample size and lack of grouping metadata make split-level attribution estimates unstable and pedagogically uninterpretable." (behavioraudit.tex, line 321)

2. **Profile-specific language.** In the interpretive paragraph, the previous wording "on Strong data, feature-attribution is stable; on Fragile or Partial data, part of the iid signal arises from split-specific noise" has been revised to "on OULAD (Strong), feature-attribution is stable; on UCI Student (Mostly Passing) and Higher Ed (Partial), part of the IID signal appears to arise from split-specific noise rather than stable underlying structure". (behavioraudit.tex, line 325)

3. **Cross-model comparison.** The previous claim that "attribution rankings from fragile datasets are model-dependent as well as split-dependent" has been revised to "attribution rankings from the Partial

and Mostly Passing datasets are more model-dependent as well as split-dependent". (behavioraudit.tex, line 325)

4. **Concluding summary.** The previous concluding sentence contrasted "Strong datasets" with "Fragile datasets"; the revised version explicitly names the three analysed datasets: "OULAD (Strong) maintains stable attribution patterns across splits, while UCI Student (Mostly Passing) and Higher Ed (Partial) exhibit more split-dependent rankings." (behavioraudit.tex, line 325)

Throughout the rest of the manuscript (Methods, Results, Discussion, Conclusion), references to "Fragile datasets" are retained because they describe the full seven-dataset audit, which does include the Fragile-profile MM-TBA. The scope restriction applies only to the feature-attribution section.

We hope these revisions fully address Reviewer 2's remaining concerns. We sincerely appreciate the reviewer's careful reading over four rounds of review, which has substantially sharpened the precision of our claims and the scope of our conclusions. We believe the manuscript now provides a more rigorous and appropriately scoped presentation of the audit protocol and its supporting analyses, and we respectfully submit it for the Editor's final consideration.

Respectfully,
Yan Ma Lizhuo Zhang